MILITARY SPECIFICATION

FUZE, PIBD, M740, FUZE MODULE FOR

This specification is approved for use by US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the Fuze Module Assembly critical item.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-60 - Tape, Pressure-Sensitive Adhesive, Waterproof, For Packing
PPP-B-601 - Box, Wood, Plywood
PPP-F-320 - Fiberboard, Corrugated and Solid

MILITARY

MIL-B-81705 - Barrier Materials, Flexible, Heat Sealable

FSC 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-100 - Engineering Drawing Practices
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-810 - Environmental Test Methods
MIL-STD-1171 - Acceptance and Description Sheets

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US Army Armament Research and Development Center (ARDC)

9307468 - Fuze Module Assembly, Tactical

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specified acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item definition. - The fuze module assembly consists of two independent channels, each containing a firing circuit and a safe and arming (S&A) mechanism (including explosive components). The S&A mechanism includes the escapement timer, rotor, detonator, piston actuator, and dual locking device. The S&A shall mechanically arm the fuzing system by aligning the explosive train only after being subjected to the two cannon environments in sequence within time limits; i.e., set-back and muzzle exit velocity. Each fuzing system firing circuit shall be electrically armed after mechanical arming is complete and after receipt of the arm warhead signal from the projectile guidance subsystem. Each firing circuit shall initiate a detonator, the first element of the explosive train.

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3.2 Design and construction.

3.2.1 Fuze module assembly. The fuze module assembly shall be fabricated and assembled in accordance with the requirements specified on Drawing 9307468 and all associated drawings and the applicable specifications and standards referenced thereon.

3.3 Characteristics

3.3.1 Performance

3.3.1.1 Fuze Module Assembly (9307468)

3.3.1.1.1 Visual arm of safe indicator. The fuze module assembly shall show a red background and a letter A, or green background and a letter S to indicate an "arm" or "safe" condition, respectively for each fuze channel.

3.3.1.1.2 Electrical isolation. The fuze module assembly external structure, consisting of fuze container, support plate and forward plate shall be electrically isolated from each of the fuze assemblies with a minimum resistance of 10 megohm. The two fuze assemblies shall be electrically isolated from each other with a minimum resistance of 10 megohm.

3.3.1.1.3 Electrical resistance. The electrical resistance between conductors in the flat cable of each fuze assembly, when the first conductor listed is connected to the positive meter terminal, shall be as follows:

Conductor	Resistance (Ohms)
1, 5	320K (min) to 400K (max)
2, 5	4.5 (min) to 11.2 (max)
3, 5	0 (min) to 0.5 (max)
4, 5	900K (min) after 30 sec

3.3.1.1.4 Arming. Both channels of the fuze module shall arm when the fuze module is subjected to launch accelerations equal to or greater than 1200 $\pm$ 400 g's for a period of 10 milliseconds minimum, and an electrical enabling signal is supplied to the fuze module at 0.4 $\pm$ .1 second after launch to function both piston actuators.

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3.3.1.1.5 Initiation. The fuze module explosive train shall be initiated when after arming, one or both firing circuits is charged to 26-35VDC and a fire signal with a 12 volt or greater peak is applied to the input of the circuit(s). The explosive train shall not initiate if a fire signal of 8.5 volts peak or less is applied to the input of the circuit(s).

3.3.1.1.6 Explosive output. Upon detonation, the fuze module booster shall produce a minimum dent of 0.05 inch in a steel witness block when tested in accordance to 4.5.2.1.6.

3.3.1.1.7 Sealing. The fuze module shall be sealed to prevent intrusion of the warhead explosive into the fuze module.

3.3.1.2 Fuze Assembly (9308511)

3.3.1.2.1 Contact assembly ground circuit resistance. The resistance between the detonator button, not depressed, and conductor 3 or 5 of the cable shall be 0.5 ohms, maximum.

3.3.1.3 Safing and arming assembly (9308512)

3.3.1.3.1 Arming enable. Function of the first event lever and second event lever shall remove the two rotor locks and enable the S&A to arm.

3.3.1.3.2 Fail safe. The rotor assembly shall return from a partially armed position to its initial position if the second event lever is not functioned.

3.3.1.3.3 Minimum acceleration. The g-weight shall rotate after which the first of two rotor unlocks shall occur. The first unlock shall always occur at an acceleration of 800 g's or greater for 10 milliseconds and shall never occur at an acceleration of 300 g's or less.

3.3.1.3.4 Arming time. The rotor assembly shall move from its initial safe position to the armed position and the detonator contact assembly short shall be removed in 1.5, +1.0, -0.5 seconds from release of the rotor. This arming time shall be met with the safe and arming assembly non-spinning at -25, +0, -5°F and at +145, +5, -0°F. At ambient temperature the arming time shall also be met with the safing and arming assembly spinning at 1800 ± 100 rpm about the lead hole axis.

3.3.1.3.5 Firing circuit resistance. With the rotor in the armed position, the detonator firing circuit resistance shall be between 2.0 ohms (min) to 5.5 ohms (max).

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3.3.1.4 Fuze Arming Timer (9307275).

3.3.1.4.1 Arming time. The escapement sector gear shall rotate at ambient temperature from its first event lever locked position in a simulated S&A assembly through 90 ± 2 degrees in 1.5, +1.0, -0.5 seconds at each of the seven mesh positions of the sector gear with its mating pinion while in the 1800 ± 100 rpm spin environment.

3.3.1.5 Fuze cup assembly (9307451)

3.3.1.5.1 Functional test. When the charging circuit is subjected to an input voltage of 26 to 35 VDC, the fuze cup electronics shall provide an output of not more than 0.2V peak. When the output is loaded with a 2 ohm resistor, the voltage across the 2 ohm resistor shall not exceed .020V peak. When a fire signal pulse of 8.5 ± 1 volt peak is applied to the input of the circuit, the output across the 2 ohm load shall not exceed .020V peak. There shall be an output across the 2 ohm load of greater than 6V at 10 microseconds after the initial voltage rise when a 12 ± 1 volt peak fire pulse is applied to the input of the circuit. The circuit shall be capable of triggering on input, pulses as fast as a single halfwave, 25 ± 1 KHZ sinusoidal pulse.

3.3.2 Environmental conditions

3.3.2.1 High temperatures. The fuze module assembly shall meet the performance requirements specified in 3.3.1 at temperatures up to 145°F maximum.

3.3.2.2 Low temperatures. The fuze module assembly shall meet the performance requirements specified in 3.3.1 at temperatures down to -25°F minimum.

3.3.2.3 Acceleration high-G (9000 G's). The fuze module shall meet the performance requirements specified in 3.3.1 with the exception that the acceleration specified in 3.3.1.1.4 shall be 9000 + 1500 - 300 g's instead of 1200 ± 400 g's for a period of 10 milliseconds minimum. The direction of acceleration is shown in Figure 1. The acceleration curve shall have no sharp discontinuities and shall be above 8000 g's for at least 1.5 milliseconds. The fuze shall be capable of meeting this requirement at temperatures from -25°F (min) to +145°F (max).

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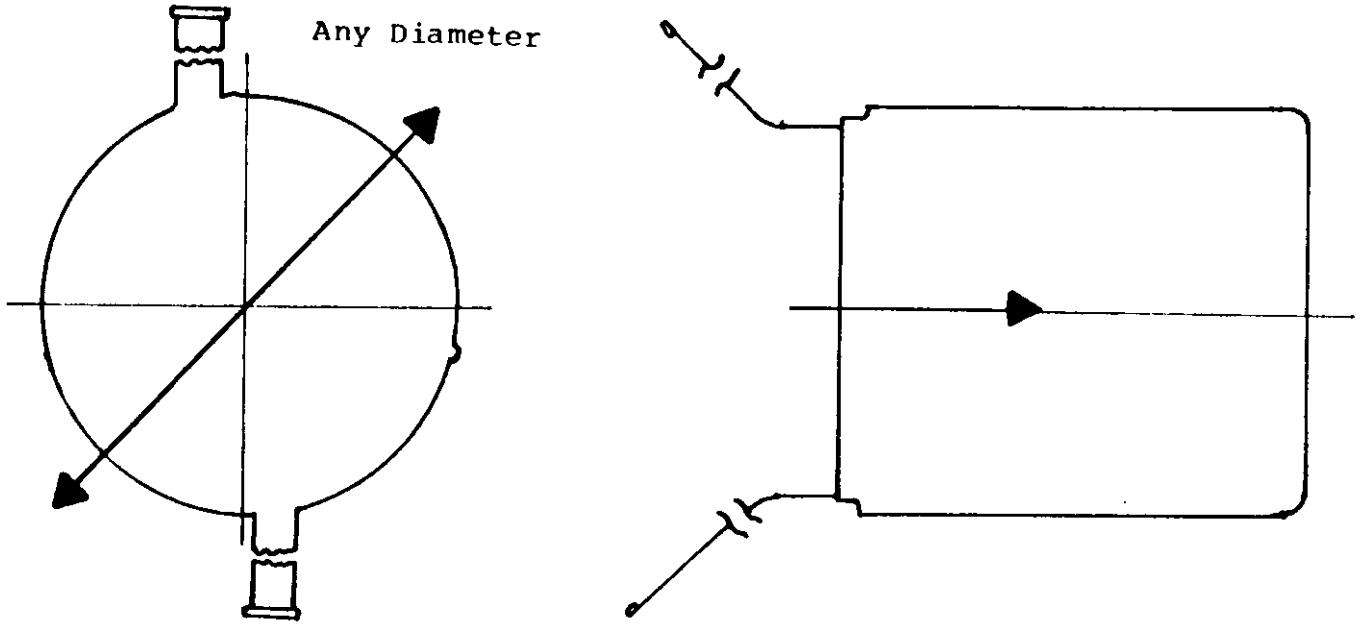


Figure 1. Direction of Applied Acceleration

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3.3.2.4 Transportation vibration. The fuze module assembly shall meet the performance requirements specified in 3.3.1 following exposure to the following vibration levels at temperatures of -50°F to 160°F.

Frequency (Hz)	Level ± 0%
5-11	0.4 inch displacement, peak to peak
12-37	2.5 g acceleration, peak
38-52	0.036 inch displacement, peak to peak
53-500	5.0 g acceleration, peak

3.3.2.5 Humidity. The fuze module assembly shall meet the performance requirements as specified in 3.3.1 following exposure to a humid atmosphere when tested as specified in 4.5.3.5.

3.3.2.6 Five-foot drop. The fuze module assembly shall meet the performance requirements as specified in 3.3.1 following a drop when tested as specified in 4.5.3.6.

3.3.2.7 Lateral acceleration. The fuze module shall meet the performance requirements specified in 3.3.1 following exposure to a minimum shock load of 750, +50, -0 g's for a period of 1 ± .1 milliseconds along the lateral axis of the fuze module assembly. The direction of the acceleration force shall be as shown in Figure 1.

3.3.2.8 Thermal shock. The fuze module assembly shall meet the performance requirements specified in 3.3.1 following exposure to thermal shock conditions when tested as specified in 4.5.3.8.

3.3.2.9 Jolt. The fuze module assembly shall not detonate and shall remain safe to handle when exposed to jolt conditions when tested as specified in 4.5.3.9.

3.3.2.10 Jumble. The fuze module assembly shall not detonate and shall remain safe to handle when exposed to jumble conditions when tested as specified in 4.5.3.10.

3.3.2.11 Forty-foot drop. The fuze module assembly shall not detonate and shall be safe to handle and dispose of following exposure to a free-fall drop when tested as specified in 4.5.3.11.

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3.4 First article inspection. This specification makes provisions for first article inspection. When specified, a sample shall be subjected to first article inspection. First article samples shall meet all the requirements of this specification. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. All parts shall be fabricated and finished in a thoroughly workmanlike manner. Except as otherwise specified, the parts shall be free of cracks, scratches, burrs, dents, gouges, surface defects or any surface discontinuities. The items shall be free from dirt, chips, grease, corrosion and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and legible. Exterior surface coatings shall be continuous. Scratches not exposing base metal are permitted. Voids in exposed potting of the fuze cup assembly not exceeding 0.25 inch wide, 0.25 inch long and 0.125 inch deep and which do not expose electrical components, leads, circuitry or the printed wiring board are permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The following types of inspections shall be conducted on this item:

a. First article inspection (4.3).

b. Quality conformance inspection (4.4).

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4.3. First article inspection.

4.3.1 Submission. The contractor, when required by contract, shall submit a first article sample of 30 fuze module assemblies, 16 lead cups, 16 booster assemblies and 5 complete sets of components and subassemblies as listed in "Classification of Defects and Tests", paragraph 4.4.2. All assemblies and components shall have been produced by the contractor or furnished by a supplier using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and material shall be obtained from the same source(s) of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to ensure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes, and tests. A sample containing known defects will not be submitted unless specifically authorized by the procuring activity.

4.3.2 Inspections to be performed. First article test assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in this specification and to any or all requirements of the applicable drawings. Environmental tests shall be conducted in the sequence indicated. The testing shall be considered destructive and the tested assemblies shall not be returned to a lot for delivery.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article samples shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or test specimen in the sample to comply with any of the stated requirements.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63232A (AR)

PARAGRAPH	TITLE: Fuze Module Assembly, Tactical		SHEET (1) 1 OF 4		DRAWING NUMBER: 9307468
					NEXT HIGHER ASSEMBLY: 9307000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:					
1	Arm or Safe Indication		100%	3.3.1.1.1	4.5.2.1.1
MAJOR:					
101	Electrical Isolation		100%	3.3.1.1.2	4.5.2.1.2
102	Electrical Resistance		100%	3.3.1.1.3	4.5.2.1.3
103	Diameter		100%	3.2	Gage
104	Width, including key		100%	3.2	Gage
105	Height, (max.)		100%	3.2	Gage
MINOR:					
201	Sealing		100%	3.3.1.1.7	4.5.2.1.4
202	Workmanship		100%	3.5	Visual
203	Marking		100%	9307468	Visual

NOTES:
(1) All tests on this page must be conducted prior to tests on sheet 2.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63232A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 4		DRAWING NUMBER
	Fuze Module Assembly, Tactical				9307468
					NEXT HIGHER ASSEMBLY
					9307000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group I - 12 Fuze Module Assemblies				
CRITICAL:					
1	Arm or Safe Indication	12	100%	3.3.1.1.1	4.5.2.1.1
MAJOR:					
106	Electrical Isolation	12	100%	3.3.1.1.2	4.5.2.1.2
107	Electrical Resistance	12	100%	3.3.1.1.3	4.5.2.1.3
108	High Temperature	12	100%	3.3.2.1	4.5.3.1
109	Low Temperature	12	100%	3.3.2.2	4.5.3.2
110	Thermal Shock	12	100%	3.3.2.8	4.5.3.8
111	Lateral Acceleration	12	100%	3.3.2.7	4.5.3.7
112	Five Foot Drop	12	100%	3.3.2.6	4.5.3.6
113	Acceleration, High-g (1)	6	100%	3.3.2.3	4.5.3.3
114	Arming (2)	6	100%	3.3.1.1.4	4.5.2.1.5
115	Initiation (3)	12	100%	3.3.1.1.5	4.5.2.1.6
116	Explosive Output (3)	12	100%	3.3.1.1.6	4.5.2.1.6
MINOR:					
201	Workmanship		0.65%	3.5	Visual

NOTES:
(1) Test 3 fuze modules per 4.5.3.3.1, test 3 fuze modules per 4.5.3.3.2.
(2) Test 3 fuze modules per 4.5.2.1.5.1, test 3 fuze modules per 4.5.2.1.5.2.
(3) Test at temperature consistent with those specified in categories 113 or 114.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63232A (AR)

PARAGRAPH	TITLE: Fuze Module Assembly, Tactical	SHEET 3 OF 4	DRAWING NUMBER: 9307468
			NEXT HIGHER ASSEMBLY: 9307000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group II - 12 Fuze Module Assemblies				
CRITICAL:					
1	Arm or Safe Indication	12	100%	3.3.1.1.1	4.5.2.1.1
MAJOR:					
117	Electrical Isolation	12	100%	3.3.1.1.2	4.5.2.1.2
118	Electrical Resistance	12	100%	3.3.1.1.3	4.5.2.1.3
119	Humidity	12	100%	3.3.2.5	4.5.3.5
120	Transportation Vibration	12	100%	3.3.2.4	4.5.3.4.1
121	Acceleration, High-g (1)	6	100%	3.3.2.3	4.5.3.3
122	Arming (2)	6	100%	3.3.1.4	4.5.2.1.5
123	Initiation (3)	12	100%	3.3.1.1.5	4.5.2.1.6
124	Explosive output (3)	12	100%	3.3.1.1.6	4.5.2.1.6
MINOR:					
201	Workmanship		0.65%	3.5	Visual

NOTES:
(1) Test 3 fuze modules per 4.5.3.3.1, test 3 fuze modules per 4.5.3.3.2.
(2) Test 3 fuze modules per 4.5.2.1.5.1, test 3 fuze modules per 4.5.2.1.5.2.
(3) Test at temperature consistent with those specified in categories 121 or 122.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63232A (AR)

PARAGRAPH	TITLE: Fuze Module Assembly, Tactical		SHEET 4 OF 4		DRAWING NUMBER: 9307468
					NEXT HIGHER ASSEMBLY: 9307000
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group III - 6 Fuze Module Assemblies				
CRITICAL:					
1	Arm or Safe Indication	6	100%	3.3.1.1.1	4.5.2.1.1
2	Jolt	2	100%	3.3.2.9	4.5.3.9
3	Jumble	2	100%	3.3.2.10	4.5.3.10
4	Forty Foot Drop	2	100%	3.3.2.11	4.5.3.11
MAJOR:					
125	Electrical Isolation (1)	6	100%	3.3.1.1.2	4.5.2.1.2
126	Electrical Resistance (1)	6	100%	3.3.1.1.3	4.5.2.1.3
MINOR:					
201	Workmanship		0.65%	3.5	Visual

NOTES:

(1) Conduct these tests before Jolt, Jumble, or Forty Foot Drop.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63232A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Lead Cup Assembly	1 OF 1	9307249
			NEXT HIGHER ASSEMBLY 9307468

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR: 101	Operational Test, minimum dent (1)		100%	3.2	Gage
MINOR: 201	Workmanship		0.65%	3.5	Visual

NOTES:
(1) One half the sample to be actuated when stablized at 145° ± 5°F.
One half the sample to be actuated when stablized at -25° ± 5°F.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63232A (AR)

PARAGRAPH	TITLE: Booster Assembly		SHEET 1 OF 1		DRAWING NUMBER: 9307251
					NEXT HIGHER ASSEMBLY: 9307468
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR: 101	Operational Test, minimum dent (1)		100%	3.2	Gage
MINOR: 201	Workmanship		0.65%	3.2	Visual

NOTES:
(1) One half the sample to be actuated when stablized at 145° ± 5°F.
One half the sample to be actuated when stablized at -25° ± 5°F.

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4.4. Quality conformance inspection

4.4.1 Inspection lot. Unless otherwise specified herein or by the procuring activity, inspection lot size, lot formation, and presentation of lots shall be in accordance with the inspection and sampling procedures herein and the "submission of product" provisions of MIL-STD-105. Lot size shall be not greater than 1 month's production. Each lot of fuze module assemblies shall contain leads, boosters, detonators, and piston actuators from not more than two lot numbers of one interfix number from one manufacturer. All other parts and subassemblies shall each be of one dash number of the same drawing number.

4.4.2 Inspections

Sampling plans. Unless otherwise specified, sampling shall be to level II of MIL-STD-105.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63232A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Detonator and Contact Assembly	1 OF 1	9307223
			NEXT HIGHER ASSEMBLY: 9308518

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Resistance between contact and housing		100%	3.2	Gage
102	Resistance between contact and housing when contact is depressed		100%	3.2	Gage
103	Load required to deflect contact		0.40%	3.2	Gage
MINOR:					
201	Workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63232A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Timer, Arming Fuze	1 OF 1	9307275
			NEXT HIGHER ASSEMBLY: 9308512

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Static arming time		100%	3.2	Note 4 of Dwg.
102	Seven position arming time		100%	3.3.1.4.1	4.5.2.4.1
MINOR:					
201	Stake separation force		0.65%	3.2	Note 2 of Dwg.
202	Workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63232A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Fuze Cup Assembly, Tactical	1 OF 1	9307451
			NEXT HIGHER ASSEMBLY: 9307455

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR: 101	Functional Test		100%	3.3.1.5.1	4.5.2.5.1, 4.5.2.5.2
MINOR: 201	Resistance between rivet nut and cable wire 3		100%	3.2	Ohm Meter
202	Workmanship - Cracks in potting		100%	3.5	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63232A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Piston Actuator and Contact Assembly	1 OF 1	9307453
			NEXT HIGHER ASSEMBLY: 9307471

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR: 101	Test force on contacts		0.40%	3.2	Gage
MINOR: 201	Workmanship		0.65%	3.5	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9307473
4.4.2.5	Base Plate Assembly				NEXT HIGHER ASSEMBLY 9307468
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR: 101	Force to push out key pin (min.)		0.40%	3.2	Gage
MINOR: 201	Workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Fuze Assembly	1 OF 1	9308511
			NEXT HIGHER ASSEMBLY: 9307468

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR: 101	Contact assembly ground circuit resistance		100%	3.3.1.2.1	4.5.2.2.1
MINOR: 201	Workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63232A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Safing and Arming Assembly		9308512
			NEXT HIGHER ASSEMBLY 9308511

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:					
1	Arming time (if less than minimum)		100%	3.3.1.3.4	4.5.2.3.4
MAJOR:					
101	Arming enable		100%	3.3.1.3.1	4.5.2.3.1
102	Fail safe		100%	3.3.1.3.2	4.5.2.3.2
103	Minimum acceleration		100%	3.3.1.3.3	4.5.2.3.3
104	Arming time (if greater than maximum)		100%	3.3.1.3.4	4.5.2.3.4
105	Firing circuit resistance		100%	3.3.1.3.5	4.5.2.3.5
MINOR:					
201	Workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Rotor Assembly	1 OF 1	9308518
			NEXT HIGHER ASSEMBLY: 9308512

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Force to move contact		0.40%	9308518 Note 4	Gage
102 (1)	Resistance from seated contact button to rotor contacts		100%	9308518 Note 8	Gage
103	Detonator contact assembly stake distortion		100%	9308518 Note 3	Visual, Gage
104	Detonator contact assembly stake unseating		0.40%	9308518 Note 5	Gage
105	Rotor lock pin stake		0.40%	9308518 Note 2	Gage
MINOR:					
201	Workmanship		0.65%	3.5	Visual

NOTES:
(1) This test must be conducted after test 101.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Fuze Module Assembly, Tactical	1 OF 1	9307468
			NEXT HIGHER ASSEMBLY 9307000

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:					
1	Arm or safe indication		100%	3.3.1.1.1	4.5.2.1.1
MAJOR:					
101	Electrical isolation		100%	3.3.1.1.2	4.5.2.1.2
102	Electrical resistance		100%	3.3.1.1.3	4.5.2.1.3
103	Diameter		0.40%	3.2	Gage
104	Width, including key		0.40%	3.2	Gage
105	Height, (max)		0.40%	3.2	Gage
106	Thermal Shock	(1)	100%	3.3.2.8	4.5.3.8
107	Transportation, Vibration	(1)	100%	3.3.2.4	4.5.3.4.2
108	High-g, High Temperature, Arming	(2)	100%	3.3.2.3	4.5.3.3.1
109	High-g, Low Temperature, Arming	(2)	100%	3.3.2.3	4.5.3.3.2
110	Low-g, High Temperature, Arming	(2)	100%	3.3.1.1.4	4.5.3.3.3
111	Low-g, Low Temperature, Arming	(2)	100%	3.3.1.1.4	4.5.3.3.4
112	Initiation (3)	(1)	100%	3.3.1.1.5	4.5.2.1.6
113	Explosive Output (3)	(1)	100%	3.3.1.1.6	4.5.2.1.6
MINOR:					
201	Sealing		0.65%	3.3.1.1.7	4.5.2.1.4
202	Workmanship		0.65%	3.5	Visual
203	Marking		0.65%	9307468	Visual

NOTES: (1) All units in the sample (ref. 4.4.3.1)
(2) One fourth each of the units in the sample (ref. 4.4.3.1)
(3) Test at temperatures consistent with those specified in category 108-111.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Lead Cup Assembly	1 OF 1	9307249
			NEXT HIGHER ASSEMBLY 9307468

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR: 101	Operational Test, minimum dent (1)	(2)	100%	3.2	Gage
MINOR: 201	Workmanship		0.65%	3.5	Visual

NOTES: (1) One half the sample to be actuated within one minute after stabilization @ 145° +5°F
One half the sample to be actuated within one minute after stabilization @ -25° ±5°F
(2) Sample quantities per 4.4.3.2.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63232A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Booster Assembly				9307251
					NEXT HIGHER ASSEMBLY 9307468
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR: 101	Operational Test, minimum dent (1)	(2)	100%	3.2	Gage
MINOR: 201	Workmanship		0.65%	3.5	Visual

NOTES: (1) One half the sample to be actuated within one minute after stabilization @ 145° +5°F
One half the sample to be actuated within one minute after stabilization @ -25° +5°F
(2) Sample quantities per 4.4.3.2.

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4.3 Testing.

4.4.3.1 Initial sampling for fuze module assembly. Beginning with the first lot produced and continuing until three consecutive lots have been found acceptable, a sample of thirty-two (32) fuze module assemblies shall be randomly selected from each lot. If two (2) or more fuze module assemblies are defective the lot shall be rejected. If one (1) or more fuze module assemblies are defective in both fuze channels (i.e., both S&A mechanisms fail to arm when tested in accordance with 4.5.2.1.5) the lot shall be rejected.

4.4.3.1.1 Subsequent sampling procedure for fuze module assembly. After three lots have complied with the criteria of 4.4.3.3, the sample size shall be sixteen (16). If two (2) or more assemblies are classified defective the lot shall be rejected. If one fuze module assembly is defective in both fuze channels, the lot shall be rejected. If one assembly is defective in only one fuze channel, a second sample of sixteen (16) assemblies shall be tested and the lot rejected if any additonal failures are found. In the event of lot rejection the sampling plan for the next lot shall revert to that of 4.4.3.1 and continue until three (3) consecutive lots have again been accepted.

4.4.3.2 Initial sampling procedure for lead cup and booster assemblies. Beginning with the first lot produced and continuing until three consecutive lots have been found acceptable, a sample of fifty (50) assemblies shall be randomly selected from each lot. If two (2) or more assemblies are classified defective, the lot shall be rejected.

4.4.3.2.1 Subsequent sampling procedure for lead cup and booster assemblies. After three consecutive lots have complied with the criteria of 4.4.3.2, the sample size shall be twenty-five (25). If two (2) or more assemblies are classified defective, the lot shall be rejected. If one (1) assembly is classified defective, a second sample of twenty-five (25) assemblies shall be tested and the lot rejected if any additional failures are found. In the event of lot rejection the sampling plan for the next lot shall revert to that of 4.4.3.2 and continue until three consecutive lots have again been accepted.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein are described in the "Paragraph Reference/Inspection Method" column in the inspection tables included herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.4.

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4.5 Methods of inspection.

4.5.1 Test conditions. Unless otherwise noted, the shorting bar shall be in place across the exposed conductors of both fuze cables during all fuze module assembly testing.

4.5.1.1 Environmental conditions. Unless otherwise specified for the purpose of test, testing shall be conducted at the following conditions:

a.	Temperature	$72^{o} \pm 18^{o}F$ ($22.2^{o} \pm 10^{o}C$)
b.	Relative Humidity	90 percent maximum
c.	Barometric Pressure	28 to 32 inches of mercury

4.5.1.2 Test tolerances. All tolerances specified herein are absolute with no allowance made for measuring equipment inaccuracy. Measurements shall ensure that the values measured are within the tolerances or limits specified.

4.5.2 Performance

4.5.2.1 Fuze module assembly

4.5.2.1.1 Arm or safe indicator. The S&A condition of safe shall be verified by confirming that a green color and letter S is visible through each window. With the exception of the condition following a proper arming environment per 4.5.2.1.5 or 4.5.7.3, a red color and a letter A is not permissible and rejects the fuze module assembly.

4.5.2.1.2 Electrical isolation. The electrical isolation shall be verified by connecting an ohmeter across each shorting bar and either of the two non-isolated (not isolated by pushing through support plate) mounting screws prior to sealing the screw head and across the two shorting bars. The ohmeter readings shall be 10 megohm or greater. The test current of the measuring instrumentation shall be limited to one milliampere, maximum.

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4.5.2.1.3 Electrical resistance. With the shorting bar removed, the resistance between electrical conductors of each cable shall be specified in 3.3.1.1.3. The test current of measuring instrumentation shall be limited to one milliampere, maximum.

4.5.2.1.4 Sealing. The fuze module assembly shall be leak tested by pressurizing the module interior to 14.5 psi minimum with dry nitrogen through one S&A mounting screw hole. The fuze module assembly shall be immersed in Freon to cover the top of the module by 1 to 5 inches. The nitrogen leak rate shall not produce a visual stream of bubbles. Leaks through the cables shall not be cause for rejection. The S&A mounting screw hole used for leak testing shall be sealed after completion of testing.

4.5.2.1.5 Fuze module assembly arming

4.5.2.1.5.1 Fuze module assembly arming, high temperature. The fuze module assembly with the shorting bars removed shall be mounted in a carrier with test circuitry which supplies an electrical enabling signal at 0.4 ± 0.1 seconds after setback to function the piston actuators. The carrier/fuze module assembly shall then be conditioned at 145, +5, -0°F for a minimum of one hour. Within 10 minutes after removal from the temperature chamber subject the carrier/fuze module assembly using a test projectile to an acceleration of 1200 ± 400 g's for a period of 10 milliseconds, minimum in the direction as shown in Figure 1. Remove the armed fuze module assembly from the test/simulated projectile and verify that both S&A assemblies are armed in accordance with 3.3.1.1.1. Place the armed fuze in a safety container and subject it to the test specified in 4.5.2.1.6.1.

4.5.2.1.5.2 Fuze module assembly arming, low temperature. The fuze module assembly with the shorting bars removed shall be mounted in a carrier with test circuitry which supplies an electrical enabling signal at 0.4 ± 0.1 seconds after setback to fire the piston actuators. The carrier/fuze module assembly shall then be conditioned at -25 +0, -5°F for a minimum of one hour. Within 10 minutes after removal from the temperature chamber subject the carrier/fuze module using a test projectile to an acceleration of 1200 ± 400 g's for a period of 10 milliseconds, minimum in the direction as shown in Figure 1. Remove the armed fuze module assembly from the test/simulated projectile and verify that both S&A assemblies are armed in accordance with 3.3.1.1.1. Place the armed fuze in a safety container and subject it to the test as specified in 4.5.2.1.6.2.

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4.5.2.1.6 Fuze module assembly initiation and explosive output

4.5.2.1.6.1 Fuze module assembly initiation and explosive output, high temperature. After conducting the test specified by 4.5.2.1.5, place the safety container with the armed fuze in a temperature chamber and condition the unit at 145 +5, -0°F for one hour, minimum. Remove the safety container from the temperature chamber. Remove the armed fuze module from the safety container and within two minutes conduct the following test: with the shorting bars removed connect one channel of the armed fuze module to test circuitry which supplies 26-35 VDC applied positive to conductor 4 and negative to conductor 5 of the flat cable. Effort should be made to alternate selection of the fuze channel tested between the forward and the aft channel during testing of more than one fuze module. Apply a no fire pulse of a single half cycle 1 ± .1 Khz sinusoidal 8.5 ± 1.0 volt peak to the selected fuze channel with positive to conductor 1 and negative to conductor 5 of the flat cable. The fuze module shall not detonate. A "firing pulse" of a single, half cycle 25 ± 1 Khz sinusoidal 12 ± 1 volt peak shall then be applied positive to conductor 1 and negative to conductor 5 of the flat cable. The fuze module shall explode. The exploded fuze module shall produce a minimum dent of 0.05 inches in a steel witness block per MIL-STD-331, Test 301.1. The steel block shall be 2 1/2 x 2 1/2 x 1 1/2/ inches. In the event the fuze module does not detonate, it shall be considered a failed assembly with one defective fuze channel and the test shall be repeated on the other fuze channel. In the event the second channel also fails to detonate, the fuze module shall be considered a failed assembly with two defective fuze channels.

4.5.2.1.6.2 Fuze module assembly initiation and explosive output, low temperature. After conducting the test specified by 4.5.2.1.5, place the safety container with the armed fuze in a temperature chamber and condition the unit at -25 +0, -5°F for one hour, minimum. Remove the safety container from the temperature chamber. Remove the armed fuze module from the safety container and within two minutes conduct the following test: with the shorting bars removed connect one channel of the armed fuze module assembly to test circuitry which supplies 26-35 VDC applied positive to conductor 4 and negative to conductor 5 of the flat cable. Effort should be made to alternate selection of the fuze channel tested between the forward and the aft channel during testing of more than one fuze module. Apply a no fire pulse of a single half cycle 1 ± .1 Khz sinusoidol 8.5 ± 1.0 volt peak to the selected fuze channel with positive to conductor 1 and negative to conductor 5 of the flat cable. The fuze module shall not detonate. A "firing pulse" of a

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single, half cycle 25 ± 1 Khz sinusoidal 12 ± 1 volt peak shall then be applied positive to conductor 1 and negative to conductors of the flat cable. The fuze module shall explode. The exploded fuze module shall produce a minimum dent of 0.05 inches in a steel witness block per MIL-STD-331, Test 301.1. The steel block shall be 2 1/2 x 2 1/2 x 1 1/2 inches. In the event the fuze module does not detonate, it shall be considered a failed assembly with one defective fuze channel and the test shall be repeated on the other fuze channel. In the event the second channel also fails to detonate, the fuze module shall be considered a failed assembly with two defective fuze channels.

4.5.2.2 Fuze assembly

4.5.2.2.1 Contact assembly ground circuit resistance. The electrical resistance between the detonator button, not depressed, and conductor 3 or 5 of the cable shall not exceed 0.5 ohms. The test current of the measuring instrumentation shall be limited to one milliampere, maximum.

4.5.2.3 S&A assembly

4.5.2.3.1 Arming enable. An external lock shall be placed on the rotor assembly to prevent rotor movement. The G-weight shall be rotated until it is locked in its' aft position by the first event lever. The second event lever shall be actuated to move the G-weight and hold the G-weight in the full evented position. Release the rotor and observe that the rotor assembly rotates to the armed position. The arming enable test may be performed after temperature conditioning of the S&A at -25°F +0, -5°F for 30 minutes and combined with the cold temperature arming time test per 4.5.2.3.4.

4.5.2.3.2 Fail safe. A force of 5 +0, -3 pounds shall be applied tangent to the surface of the rotor assembly in a direction towards the armed position. The first event lever shall be observed to lock the rotor.

The second event lever shall be positioned in the unactuated position by installation of a piston actuator holder (9302306) or a mechanism that provides equivalent positioning. Rotating the G-weight to the stop position (full evented position) and holding it in this position shall cause the release of the first event lever lock on the rotor assembly, permitting the rotor to turn until blocked by the second event lever. Release of the G-weight shall cause the entire mechanism to return to the full safe position.

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Then an external lock shall be placed on the rotor assembly to prevent rotor movement. Rotating the G-weight in the arming direction until it is latched by the first event lever shall cause the G-weight to remain latched in that position. Release of the rotor shall cause the rotor to cycle to a partially armed position and then the entire mechanism shall return to the initial safe position. The second event lever shall be positioned in the actuated position by installation of a device that provides equivalent positioning. Rotating the G-weight to the point where it is blocked by the second event lever shall not cause the rotor to turn to the fully armed position or cause the G-weight to be latched by the first event lever. Release of the G-weight shall cause the rotor to return to the full safe position and be locked by the first event lever.

4.5.2.3.3 Minimum acceleration. Externally lock the rotor assembly from rotating during the test. Apply a 300 + 50, -0g acceleration to the S&A with its base mounted outward in an appropriate centrifuge. The specified g-level shall not cause the G-weight to be locked in its aft position by the first event lever. Apply an 800g (maximum) acceleration to the S&A in the same manner as above. The specified g-level shall cause the G-weight to be locked in its aft position by the first event lever.

4.5.2.3.4 Arming time (S&A). An external lock shall be placed on the rotor to prevent rotor movement and the G-weight moved to setback position and the second event lever held to simulate piston actuator function. Upon release of the rotor, the rotor assembly shall rotate to the armed position. The static arming time shall be measured while the S&A is at $-25^{\circ}F \pm 5^{\circ}F$ following conditioning of the S&A at $-25^{\circ}F$ +0, $-5^{\circ}F$ for 30 minutes and shall be 1.5 + 1.0, -0.5 seconds. The dynamic arming time shall be measured in a spin fixture while operating at 1800 + 100 rpm under ambient temperature and shall be 1.5 + 1.0, -0.5 seconds. The static arming time shall be measured while the S&A is at $+145^{\circ}F + 5^{\circ}F$ following conditioning of the S&A at $145^{\circ}F$ +5, $-0^{\circ}F$ for 30 minutes and shall be 1.5 +1.0, -0.5 seconds.

4.5.2.3.5 Firing circuit resistance. The rotor shall be moved to the armed position and the resistance between eyelets shall be measured. The resistance shall be from 2.0 ohms to 5.5 ohms. The test current of measuring instrumentation shall be limited to 1.0 milliampere, maximum.

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4.5.2.4 Fuze arming timer

4.5.2.4.1 Arming time. With the rotor released as in 4.5.2.3.4, the timer shall meet the 1.5 + 1.0, -0.5 second arming time for each of the seven mesh positions of the sector gear with its mating pinion while mounted in a simulated S&A test fixture and rotating at 1800 ± 100 rpm.

4.5.2.5 Fuze cup assembly functional tests

4.5.2.5.1 No fire. Functionally test the assembly by applying 35 + 1, -0 VDC, positive to wire 4, negative to wire 5 of the flat cable. The output voltage across the detonator leads shall be less than 0.2V peak. Load the detonator leads with a 2.0 ± 0.3 ohm resistor. The output across the 2 ohm resistor shall not exceed .020V peak. Apply a single half cycle 1 ± 0.1 Khz sinusoidal pulse of an amplitude of 8.5 ± 1.0 volt peak positive to wire 1, negative to wire 5 of the flat cable. The output across the two ohm resistor shall not exceed .020V peak.

4.5.2.5.2 All fire. Functionally test the assembly by applying 26 + 0, -1 VDC, positive to wire 4, negative to wire 5 of the flat cable. The output voltage across the detonator leads shall be less than 0.2V peak. Load the detonator leads with a 2.0 ± 0.3 ohm resistor. The output across the 2 ohm resistor shall not exceed .020V peak. Apply a single half cycle 2.5 ± 1 Khz sinusoidal pulse of an amplitude pf 12 ± 1.0 volt peak, positive to wire 1, negative to wire 5 of the flat cycle. The output across the 2 ohm resistor shall be greater than 6 volts at 10 microseconds after the initial voltage rise.

4.5.3 Environmental tests

4.5.3.1 High temperature. The fuze module assembly shall be conditioned at 145 +5, -0°F for a minimum of one hour. Remove the fuze module from the temperature chamber and within two minutes test both channels of the fuze module in accordance with 4.5.2.1.1 through 4.5.2.1.3.

4.5.3.2 Low temperature. The fuze module shall be conditioned at -25 +0, -5°F for a minimum of one hour. Remove the fuze module from the temperature chamber and within two minutes test both channels of the fuze module in accordance with 4.5.2.1.1 through 4.5.2.1.3.

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4.5.3.3 Acceleration - High-G (9000g's) and Low-G (1200 g's).

4.5.3.3.1 Acceleration - High-G (9000g's), high temperature. The fuze module shall be tested in accordance with 4.5.2.1.5.1 with the exception that the acceleration shall be 9000 + 1500,-300g's instead of 1200 ± 400g's for a period of 10 milliseconds, minimum. The direction of acceleration is shown in Fig. 1. The acceleration curve shall have no sharp discontinuities and shall be above 8000g's for at least 1.5 milliseconds.

4.5.3.3.2 Acceleration - High-G (9000g's), low temperature. The fuze module shall be tested in accordance with 4.5.2.1.5.2 with the exception that the acceleration shall be 9000 + 1500, -300g's instead of 1200 ± 400g's for a period of 10 milliseconds, minimum. The direction of acceleration is shown in Fig 1. The acceleration curve shall have no sharp discontinuities and shall be above 8000g's for at least 1.5 milliseconds.

4.5.3.3.3 Acceleration (1200 g) high temperature. Test in accordance with 4.5.2.1.5.1.

4.5.3.3.4 Acceleration (1200 g) low temperature. Test in accordance with 4.5.2.1.5.2.

4.5.3.4 Transportation vibration.

4.5.3.4.1 First article. The fuze module assembly shall be tested as specified in 4.5.2.1.1 through 4.5.2.1.3 after completion of transportation vibration cycling. The fuze module assembly shall be tested as specified in MIL-STD-331, Test 119, Procedure 1, with the following exceptions:

a. The cold temperature shall be -50°F rather than -65°F.

b. The stabilization time shall be 1 hour rather than 4 hours.

4.5.3.4.2 Quality conformance. The fuze module assembly shall be tested as specified in MIL-STD-331, Test 119, Procedure 2, except the units shall be vibrated in each axis for 30 minutes with sweep time of 15 minutes.

4.5.3.5 Humidity. The fuze module assembly shall be tested as specified in 4.5.2.1.1 through 4.5.2.1.3 after being subjected to the humidity environments in accordance with MIL-STD-810, Method 507.1, Procedure II.

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4.5.3.6 Five foot drop. The fuze module assembly shall be tested as specified in 4.5.2.1.1 through 4.5.2.1.3 after being subjected to a five foot drop at ambient conditions in accordance with MIL-STD-331, Test 111.1, Procedure 1, Criteria B. The test vehicle shall be a 105 mm M314A3 illuminating projectile. One fuze module assembly shall be dropped in each orientation and no fuze module assembly shall be subjected to more than one drop.

4.5.3.7 Lateral acceleration. The fuze module assembly shall be tested as specified in 4.5.2.1.1 through 4.5.2.1.3 after being exposed to a minimum shock load of 750 + 50, -0 g's with a shock pulse of 1 ± .1 millisecond. The direction of the applied acceleration force shall be as shown in Figure 1.

4.5.3.8 Thermal shock. The fuze module assembly shall be safe and be tested as specified in 4.5.2.1.1 through 4.5.2.1.3 after testing in accordance with MIL-STD-810, Method 503.1 except the low temperature shall be -50°F.

4.5.3.9 Jolt. The fuze module assembly shall meet the requirements of 3.3.2.9 after testing in accordance with MIL-STD-331, Test 101.1.

4.5.3.10 Jumble. The fuze module assembly shall meet the requirements of 3.3.2.10 after testing in accordance with MIL-STD-331, Test 102.1.

4.5.3.11 Forty foot drop. The fuze assembly shall meet the requirements of 3.3.2.11 after being subjected to a forty foot drop at ambient conditions in accordance with MIL-STD-331, Test 103, Procedure 1 except that one test vehicle shall be impacted in a longitudinal axis vertical attitude with nose down and a second test vehicle shall be impacted in a longitudinal axis vertical attitude with nose up. The test vehicle shall be a 105 mm M314A3 illuminating projectile. One fuze module assembly shall be dropped in each orientation and no fuze module shall be subjected to more than one drop.

5. PACKAGING

5.1 Preservation and packaging - Level B. Flat cables on each device must be folded so that the shorting bar is located in groove in center top of the fuze module assembly. Apply tape conforming to PPP-T-60 over the shorting bars and extending a minimum of 1 1/2" down sides of the fuze module assembly. Place each item in a non-conductive bag per MIL-B-81705, Type II, and seal.

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5.2 Packing -Level B. Items are to be packed in wood box conforming to Federal Specification PPP-B-601 having 200 lb C flute corrugated fiberboard partitions per PPP-F-320 (one part per cell). The box shall be sealed by nails and a minimum of two steel straps.

5.3 Marking. Boxes shall be marked in accordance with MIL-STD-129 and the following special D.O.T. marking:

DETONATING FUZE

CLASS C EXPLOSIVE

HANDLE CAREFULLY

DO NOT STORE OR LOAD WITH ANY HIGH EXPLOSIVE

In addition one D.O.T. "EXPLOSIVE C" label shall be applied adjacent to the consignee's name and address on the shipping container.

6. NOTES

6.1 Intended use. The item described by this specification is intended for use in the Projectile, 155MM, CLGP, M712.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Serialization requirements, if applicable.

d. Quality Conformance Inspection, if other than specified in Section 4 of this specification.

e. First Article Sample requirements, if other than specified in Section 4 of this specification.

f. Packaging requirements, if other than specified in Section 5 of this specification.

g. Certificates of conformance for each lot or shipment of product.

h. Requirements for lot control.

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6.2.2 Contract data requirements.

a. Contract data requirements for inspection equipment designs (conforming to Data Item Description DI-R-1714, Tailored).

b. Data Cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410, except that for propellant or explosive lots, Acceptance and Description Sheets will be prepared in accordance with MIL-STD-1171.

6.3 Definitions.

6.3.1 Contractor. Unless otherwise specified, the word contractor used throughout this specification shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, subcontractors, Government Owned, Contractor Operated (GOCO) plants.

6.3.2 Technical data package (TDP). A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable technical data such as plans, drawings and associated lists of specifications, standards, models, performance requirements, quality assurance provisions and packaging data.

Clarification of Quality Assurance terms and definitions are available in MIL-STD-109.

6.4 Inspection equipment design. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in Section 4, or on the equipment lists referenced on the equipment tabulation, whichever is specified in the item detail specification. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.4.2 and 6.4.3 should he elect to do so.

6.4.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

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b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.4.2 Contractor design. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public). Contractor designs shall include appropriate operation instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalogue listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered.

6.4.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. All submissions shall be addressed to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-QAR-I(D), Dover, NJ 07801.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as, US Army Armament Research and Development Command (ARRADCOM) or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1390-A410)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-63232A	FUZE, PIBD, M740, FUZE MODULE FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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